

LAB 3: DESIGN SPRINT EXPERIMENT REPORT

Topic: Electricity Outage Information Is Not Communicated Clearly

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1. Aim of the Experiment

The aim of this experiment is to apply the **Design Sprint / Design Thinking methodology** by following the stages **Empathize, Define, Ideate, Prototype, and Test** to design a **user-centered solution** for people who face difficulty in receiving **clear and timely electricity outage information**.

The focus is on **understanding real user problems** and **validating solutions through feedback**, rather than making assumptions.

2. Problem Context

Electricity outages occur frequently due to **maintenance, overload, weather conditions, or technical faults**.

However, outage information is often:

- Communicated **late or not at all**
- Shared through **unclear messages**
- Available only on **websites or offices difficult to access**
- Missing **restoration time estimates**

Because of this, users experience:

- **Frustration and uncertainty**
- Difficulty planning daily activities
- Problems for **students, workers, and businesses**
- Lack of trust in electricity service providers

3. Empathize Stage

Objective

To deeply understand the **needs, emotions, behaviors, and frustrations** of users during electricity outages.

Activities Performed

- Reviewed observations and notes from earlier labs
- Spoke with **household users, students, and shop owners**
- Observed reactions when **power suddenly went off**
- Checked how users try to **find outage information**

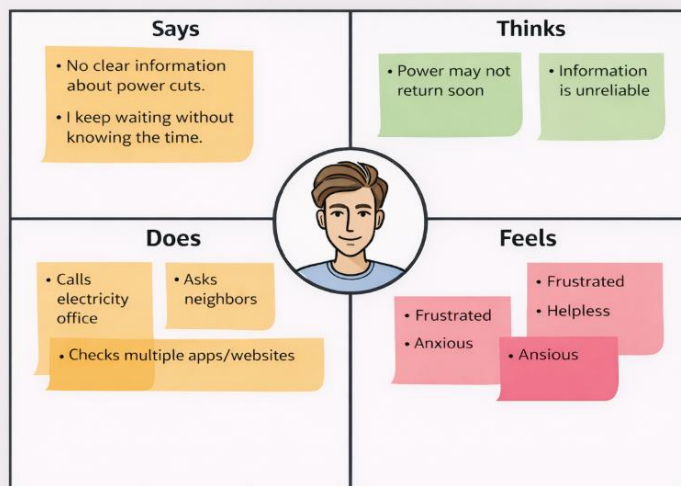
Key Observations

- Users **search multiple sources** (neighbors, WhatsApp, websites).
- No **clear restoration time** is provided.
- Messages from electricity boards are **technical and confusing**.
- Users feel **helpless and irritated**.

Empathy Map Summary



Empathy Map: Power Outage User



User Insight

Users **do not only need electricity**; they need **clear, timely, and understandable outage communication** to plan their activities confidently.

4. Define Stage

Objective

To convert user insights into a **clear and actionable problem statement**.

Problem Statement

Residents need a **simple and reliable way to receive clear electricity outage information** because the current communication is **unclear, delayed, or unavailable**, causing frustration and poor planning.

How Might We (HMW) Question

How might we provide residents with clear, real-time, and easy-to-understand electricity outage information so they can plan their daily activities confidently?

5. Ideate Stage

Objective

To generate **multiple possible solutions** without early judgment.

Brainstormed Ideas

- Mobile app with **real-time outage alerts**
- **SMS notification system** for all registered users
- **Map showing affected areas** and restoration time
- **Voice call alert** for elderly users
- WhatsApp **automatic outage chatbot**
- Color-coded **power status indicator** (Green/Red)
- Option to **report local outage quickly**
- Estimated **countdown timer for restoration**

Idea Selection

Selected Solution:

A Real-Time Electricity Outage Information Mobile System with:

- Instant alerts
- Clear restoration time
- Simple language
- Area-wise status map

Justification:

- High **user value**
- Technically **feasible**
- Easy to **understand and use**
- Works for **large population**

6. Prototype Stage

Objective

To build a **low-fidelity, testable prototype** of the selected solution.

Prototype Description

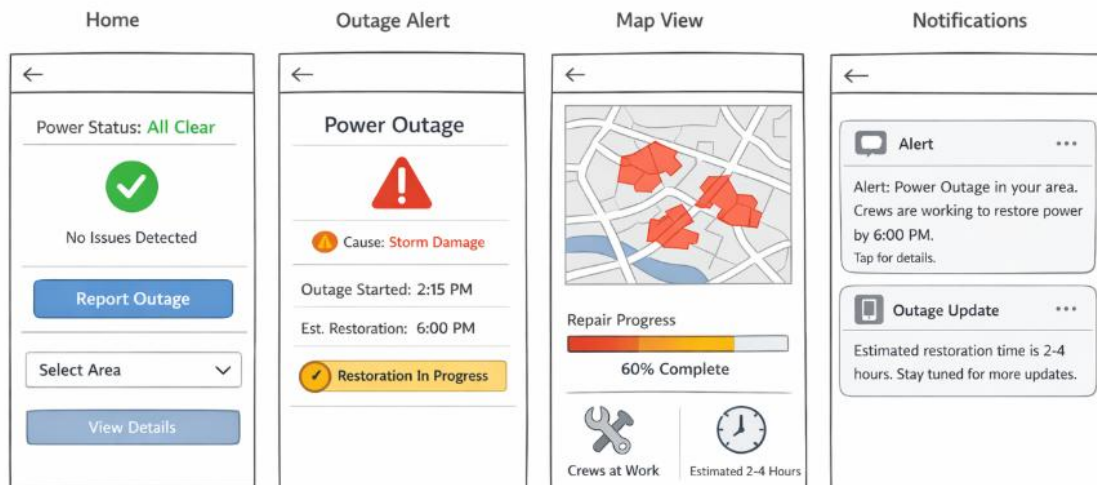
The prototype includes **simple wireframe screens**:

1. **Home Screen**
 - Current power status
 - “Report Outage” button
 - Area selection
2. **Outage Alert Screen**
 - Cause of outage
 - Start time
 - Expected restoration time
 - Simple icons and colors
3. **Map View**
 - Highlighted affected areas
 - Estimated repair progress
4. **Notification Screen**
 - SMS / app alert preview

Design Rationale

- **Simple layout** reduces confusion
- **Color indicators** improve quick understanding
- **Time estimate** reduces anxiety
- **Notifications** prevent repeated searching

Prototype Type



7. Test Stage

Objective

To evaluate the prototype using **real user interaction and feedback**.

Testing Method

- Showed prototype to **two residents and one shop owner**
- Asked them to **check outage status and restoration time**
- Observed behavior and recorded feedback

User Feedback Summary

What Worked Well

- Easy to **see outage information quickly**
- Restoration time was **very helpful**
- Map view improved **area understanding**

What Was Confusing

- Some icons needed **clear labels**
- Users wanted **regional language support**

User Suggestions

- Add **voice announcement**
- Provide **offline SMS alerts**
- Show **repair progress percentage**

Observation

Users felt **more confident and less frustrated** compared to the current real-life situation.

8. Iteration and Improvements

Based on testing:

- Add **regional language options**
- Improve **icon clarity with text labels**
- Include **voice and SMS alerts**
- Provide **live repair progress tracking**

9. Reflection

This experiment showed that the real issue was **not only power failure**, but **poor communication**.

Initial assumptions focused on **technical outage problems**, but user feedback revealed the major pain point was **lack of clear information**.

Key learnings:

- **Empathy is critical** in system design
- Small features like **time estimate and alerts** greatly reduce stress
- **User testing changes design direction**
- Communication design is as important as **technical infrastructure**

10. Conclusion

The **Design Sprint process** successfully helped identify and solve a **real user communication problem** related to electricity outages.

Through **empathy, ideation, prototyping, and testing**, a **user-friendly outage information system** was proposed.